

BSCS 6TH

LECTURE 03

TCP/IP Model

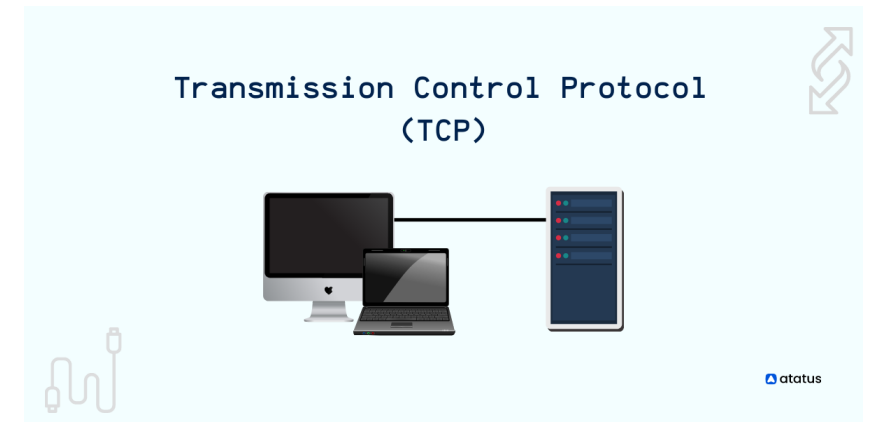
Computer Networks

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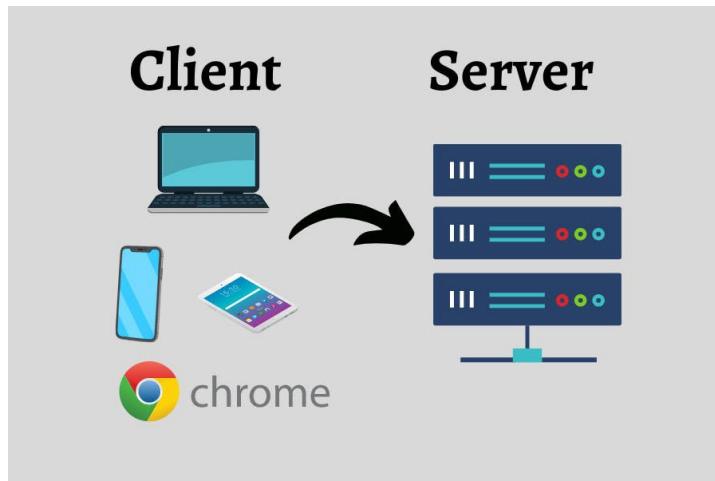
What Is TCP?

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- ▶ Transmission Control Protocol (TCP) is a communications standard that enables application programs and computing devices to exchange messages over a network. It is designed to send packets across the internet and ensure the successful delivery of data and messages over networks.
- ▶ TCP is one of the basic standards that define the rules of the internet and is included within the standards defined by the Internet Engineering Task Force (IETF). It is one of the most commonly used protocols within digital network communications and ensures end-to-end data delivery.



What Is TCP? (Cont'd)



- ▶ TCP organizes data so that it can be transmitted between a server and a client.
- ▶ It guarantees the integrity of the data being communicated over a network. Before it transmits data, TCP establishes a connection between a source and its destination, which it ensures remains live until communication begins.
- ▶ It then breaks large amounts of data into smaller packets, while ensuring data integrity is in place throughout the process.

What is UDP?

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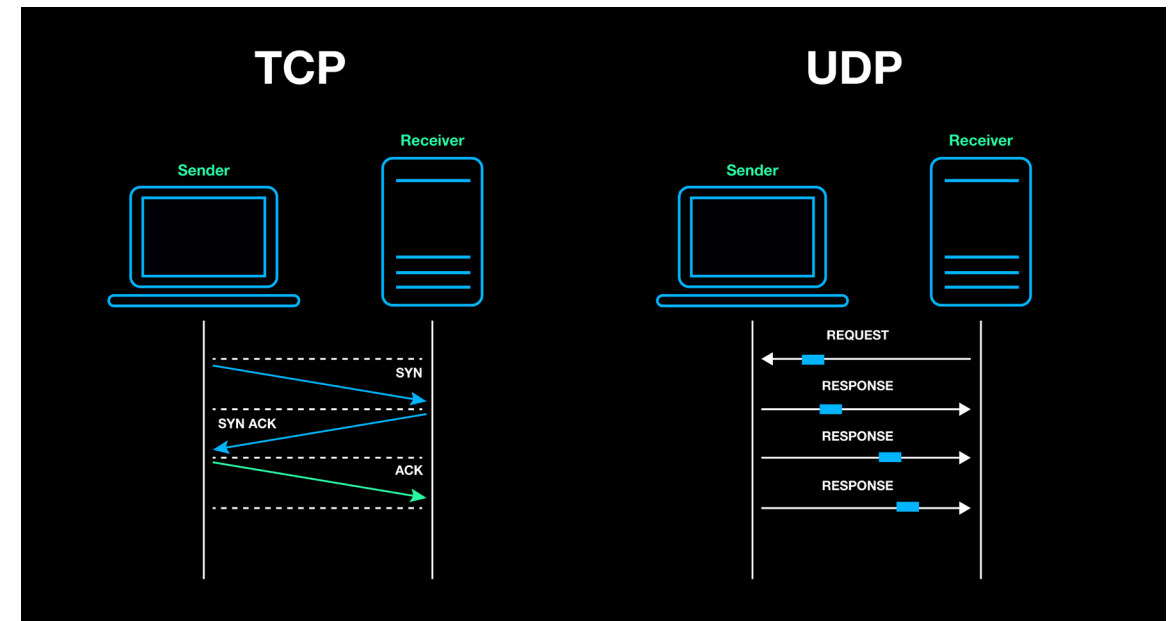


- ▶ An alternative to TCP in networking is the User Datagram Protocol (UDP), which is used to establish low-latency connections between applications and decrease transmissions time. TCP can be an expensive network tool as it includes absent or corrupted packets and protects data delivery with controls like acknowledgments, connection startup, and flow control.
- ▶ UDP does not provide error connection or packet sequencing nor does it signal a destination before it delivers data, which makes it less reliable but less expensive. As such, it is a good option for time-sensitive situations, such as Domain Name System (DNS) lookup, Voice over Internet Protocol (VoIP), and streaming media.

TCP vs UDP

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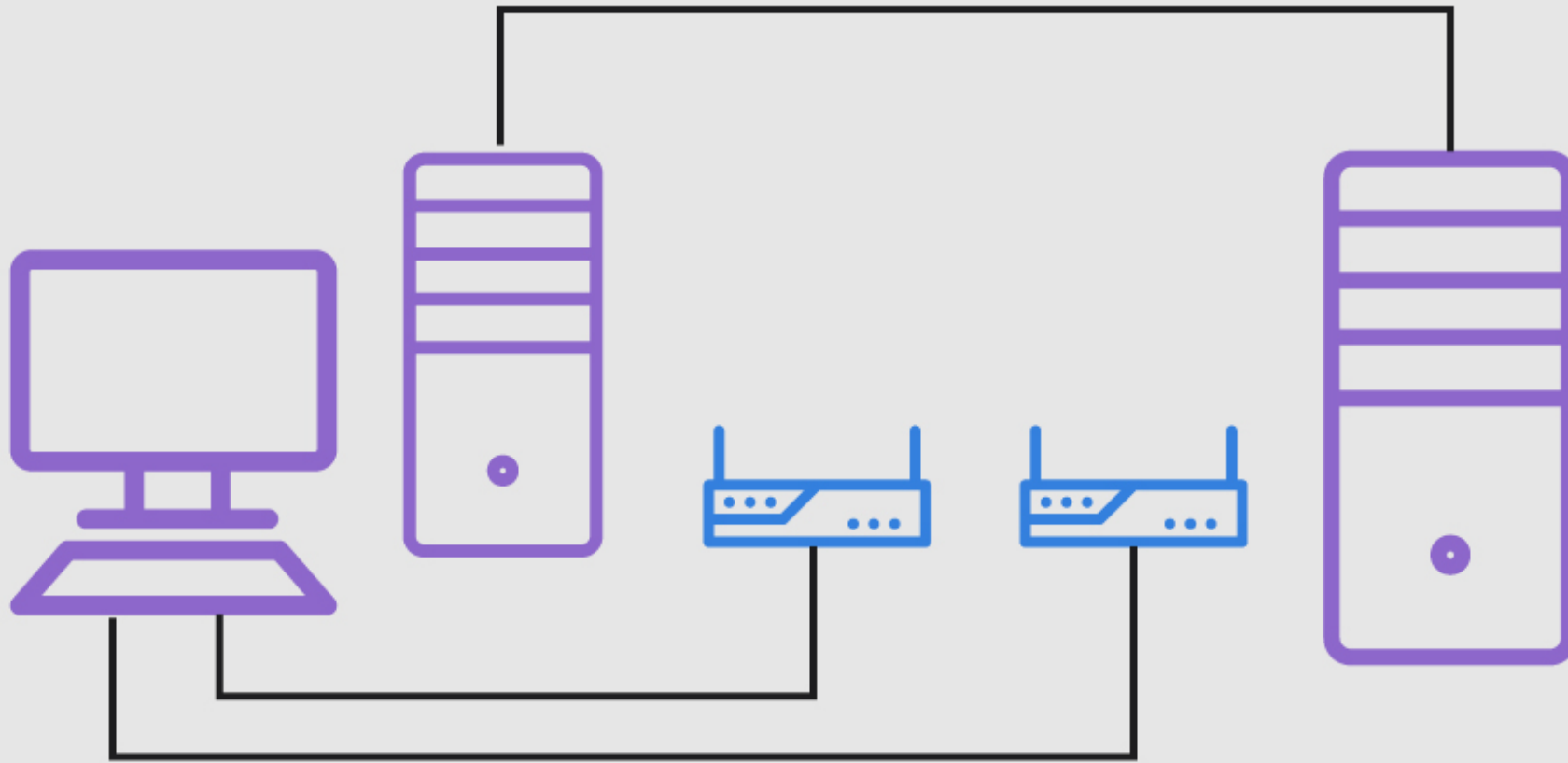
- ▶ UDP is faster but less reliable than TCP, another common transport protocol. In a TCP communication, the two computers begin by establishing a connection via an automated process called a 'handshake.' Only once this handshake has been completed will one computer actually transfer data packets to the other.
- ▶ UDP communications do not go through this process. Instead, one computer can simply begin sending data to the other.



What is IP?

- ▶ The Internet Protocol (IP) is the method for sending data from one device to another across the internet.
- ▶ Every device has an IP address that uniquely identifies it and enables it to communicate with and exchange data with other devices connected to the internet.
- ▶ Today, it's considered the standard for fast and secure communication directly between mobile devices.

What is TCP/IP?



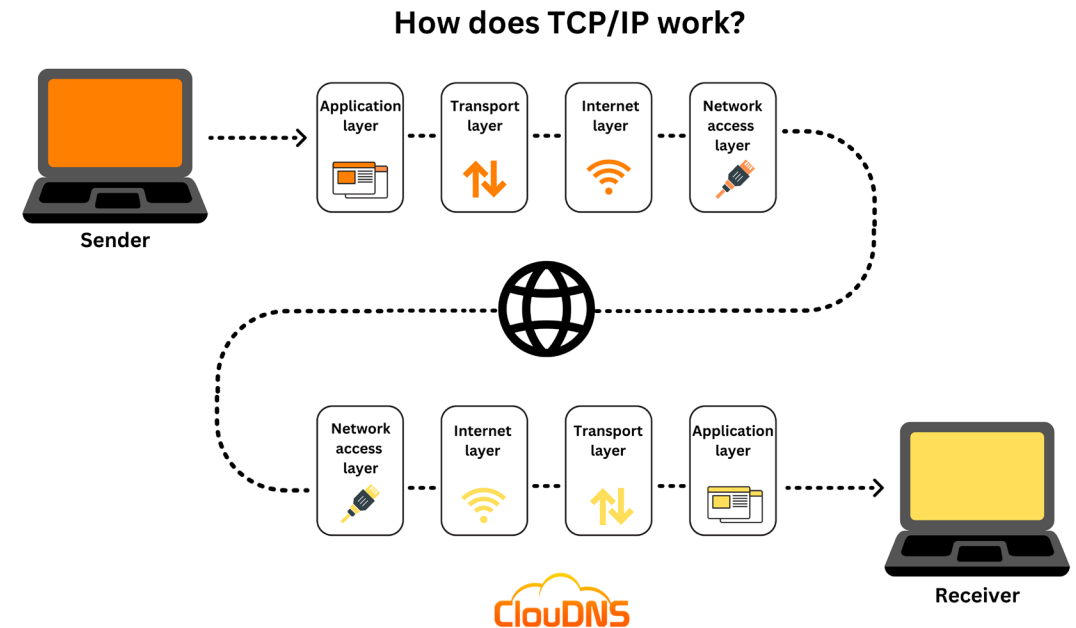
TCP/IP Model

- ▶ The TCP/IP model is the default method of data communication on the Internet.
- ▶ It was developed by the United States Department of Defense to enable the accurate and correct transmission of data between devices.
- ▶ It breaks messages into packets to avoid having to resend the entire message in case it encounters a problem during transmission.
- ▶ Packets are automatically reassembled once they reach their destination.
- ▶ Every packet can take a different route between the source and the destination computer, depending on whether the original route used becomes congested or unavailable.

How Does Transmission Control Protocol (TCP)/IP Work?

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A good example of how this works in practice is when an email is sent using SMTP from an email server. To start the process, the TCP layer in the server divides the message into packets, numbers them, and forwards them to the IP layer, which then transports each packet to the destination email server. When packets arrive, they are handed back to the TCP layer to be reassembled into the original message format and handed back to the email server, which delivers the message to a user's email inbox.



Features of the TCP/IP Model

Below mentioned are some of the features that make the TCP/IP model stand out in the network concepts:

- ▶ The TCP/IP model is among one of the most important network concepts that contributed to the working of ARPANET.
- ▶ The TCP/IP model comprises four layers: the network access layer, the internet layer, the transport layer, and the application layer (going from bottom to top).
- ▶ The network model is implemented during network and communication-related issues.
- ▶ Communication between different modes of network devices is possible through the application of various layers.
- ▶ The layers in the model provide maintenance of communication channels, flow control, and reliability check format, among other applications in the form of protocols.

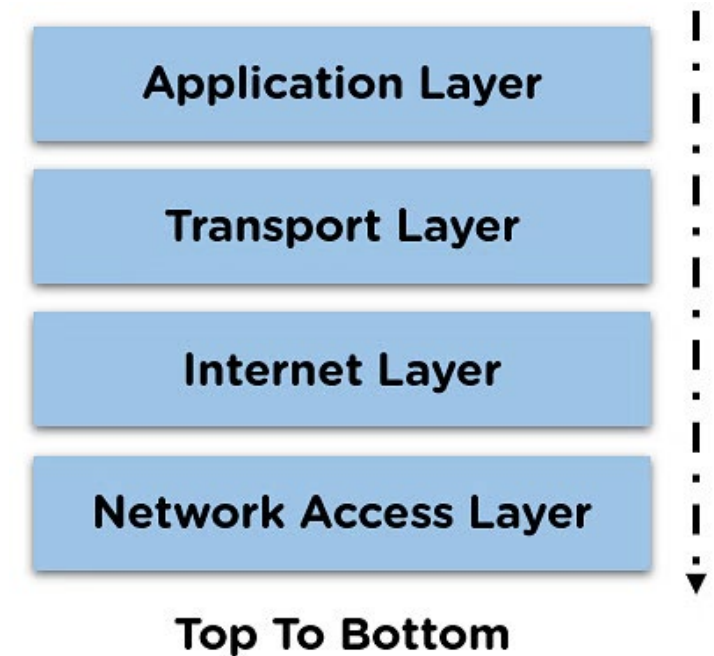
Layers of the TCP/IP Model

The TCP/IP model is divided into four different layers:

- ▶ Application layer
- ▶ Transport layer
- ▶ Internet layer
- ▶ Network Access layer

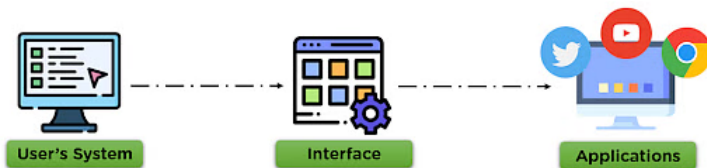
Each layer performs a specific task on the data that is being transmitted over the network channel, and data moves from one layer to another in a preset pattern as mentioned.

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Application Layer

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This is the topmost layer which indicates the applications and programs that utilize the TCP/IP model for communicating with the user through applications and various tasks performed by the layer, including data representation for the applications executed by the user and forwards it to the transport layer.

The application layer maintains a smooth connection between the application and user for data exchange and offers various features as remote handling of the system, e-mail services, etc.

Application Layer (Cont'd)

Some of the protocols used in this layer are:

- ▶ HTTP: Hypertext transfer protocol is used for accessing the information available on the internet.
- ▶ SMTP: Simple mail transfer protocol, assigned the task of handling e-mail-related steps and issues.
- ▶ FTP: This is the standard protocol that oversees the transfer of files over the network channel.



Transport Layer

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This layer is responsible for establishing the connection between the sender and the receiver device and also performs the task of dividing the data from the application layer into packets, which are then used to create sequences.

It also performs the task of maintaining the data, i.e., to be transmitted without error, and controls the data flow rate over the communication channel for smooth transmission of data.



Transport Layer (Cont'd)

The protocols used in this layer are:

- ▶ TCP: Transmission Control Protocol is responsible for the proper transmission of segments over the communication channel. It also establishes a network connection between the source and destination system.
- ▶ UDP: User Datagram Protocol is responsible for identifying errors, and other tasks during the transmission of information. UDP maintains various fields for data transmission such as:
- ▶ Source Port Address: This port is responsible for designing the application that makes up the message to be transmitted.
- ▶ Destination Port Address: This port receives the message sent from the sender side.
- ▶ Total Length: The total number of bytes of the user datagram.
- ▶ Checksum: Used for error detection of the message at the destination side.

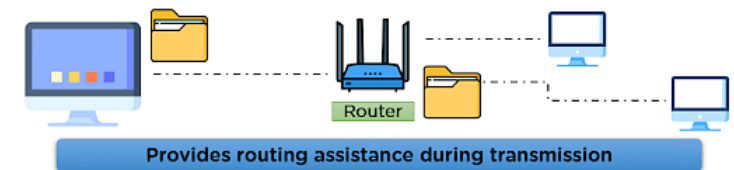
Internet Layer

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The Internet layer performs the task of controlling the transmission of the data over the network modes and enacts protocols related to the various steps related to the transmission of data over the channel, which is in the form of packets sent by the previous layer.

This layer performs many important functions in the TCP/IP model, some of which are:

- ▶ It is responsible for specifying the path that the data packets will use for transmission.
- ▶ This layer is responsible for providing IP addresses to the system for the identification matters over the network channel.



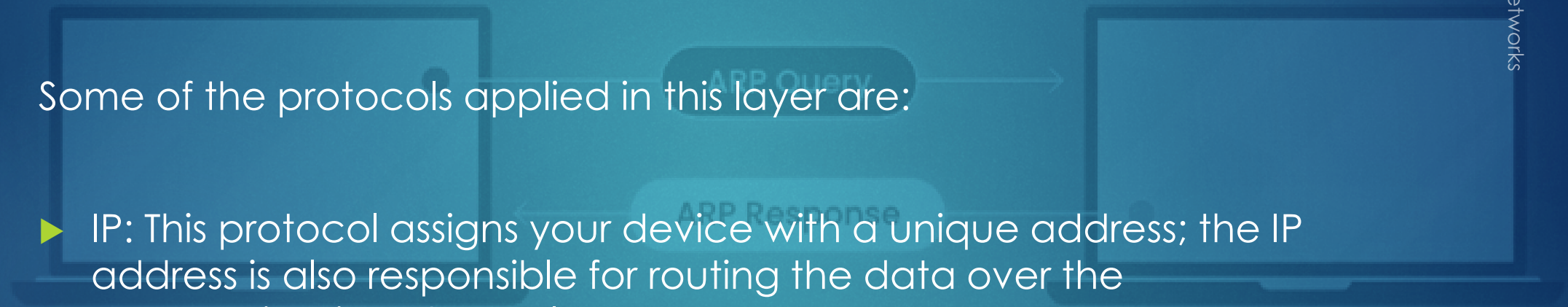
Internet Layer (Cont'd)

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Some of the protocols applied in this layer are:

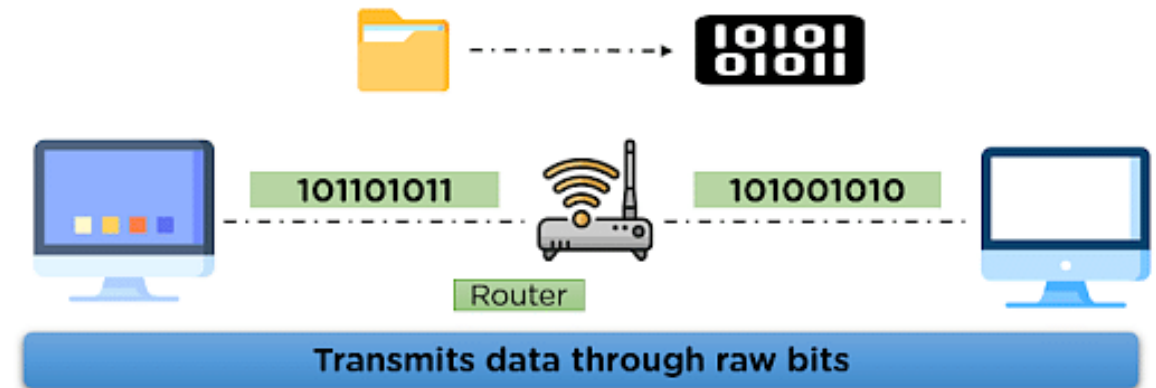
- ▶ IP: This protocol assigns your device with a unique address; the IP address is also responsible for routing the data over the communication channel.
- ▶ ARP: This protocol refers to the Address Resolution Protocol that is responsible for finding the physical address using the IP address.



Network Access Layer

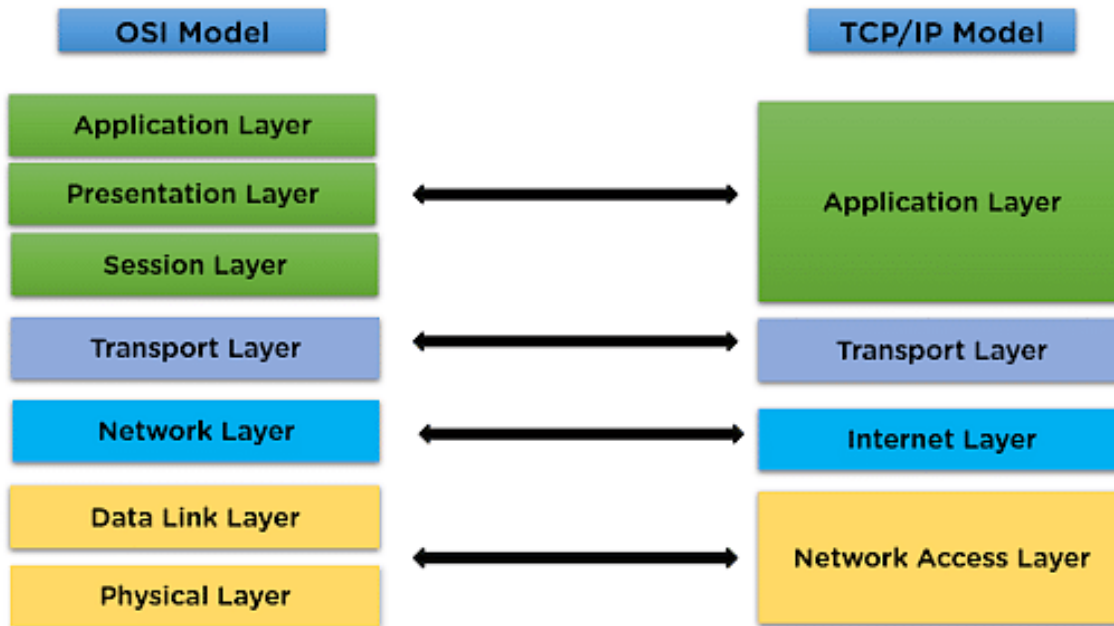
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- ▶ This layer is the combination of data-link and physical layer, where it is responsible for maintaining the task of sending and receiving data in raw bits, i.e., in binary format over the physical communication modes in the network channel.
- ▶ It uses the physical address of the system for mapping the path of transmission over the network channel.



OSI Model vs. TCP/IP Model

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The TCP/IP model was designed in the 1960s to maintain and explain the transmission of data, whereas the OSI model is a network concept specifically for explaining the communication and working of data and protocols during the transmission of information.

OSI Model	TCP/IP Model
The OSI model consists of 7 layers.	TCP/IP model comprises 4 layers.
The OSI model has separate session and presentation layers.	This model comprises a session and presentation layer in the application layer.
The transport layer in this model provides a packet delivery protocol.	In this model, the transport layer does not have any such protocols.
This model is implemented during network communication.	This model is used as a reference model for the network channel.



Thank
you